

Hypothesis Testing

- Tests a claim about a parameter using evidence (using data in a sample)
- The technique is introduced by considering a one-sample z test and t test.
- The procedure is broken into four steps
- *Each* element of the procedure must be understood

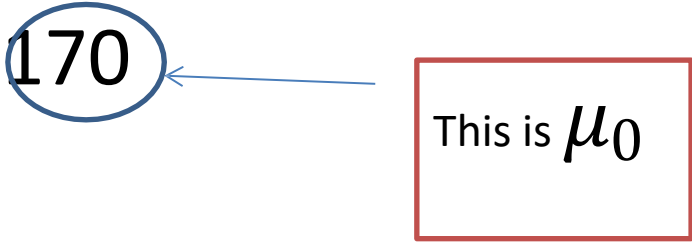
Hypothesis Testing Steps

- A. Null and alternative hypotheses
- B. Test statistic
- C. Z score or t score
- D. Conclusion

1. Null and Alternative Hypotheses

- Convert the research question to null and alternative hypotheses (compare our parameter to hypothesized value μ_0)
- The **alternative hypothesis (H_a)** formulating our question as hypothesis($<$, $>$, $\neq$)
- The **null hypothesis (H_0)** is the complement of the alternative hypothesis($\geq$, $\leq$, $=$)
- Collect data and seek evidence against H_0 as way of supporting H_a

Example: “Body Weight”

- **The problem:** In the 1970s, 20–29 year old men in the U.S. had a mean μ body weight of 170 pounds. Standard deviation σ was 40 pounds. We test whether mean body weight in the population now differs.
- **Alternative hypothesis**
 $H_a: \mu \neq 170$ (two-sided test)
- **Null hypothesis**
- $H_0: \mu = 170$ 

Example: “Body Weight”

- **The problem:** In the 1970s, 20–29 year old men in the U.S. had a mean μ body weight of 170 pounds. Standard deviation σ was 40 pounds. We test whether mean body weight in the population now is greater than 170 pounds.
- The **alternative hypothesis**
 $H_a: \mu > 170$ (**one-sided test**)
- **Null hypothesis**
 $H_0: \mu \leq 170$

Example: “Body Weight”

- **The problem:** In the 1970s, 20–29 year old men in the U.S. had a mean μ body weight of 170 pounds. Standard deviation σ was 40 pounds. We test whether mean body weight in the population now is less than 170 pounds.
- The **alternative hypothesis**
 $H_a: \mu < 170$ (**one-sided test**)
- **Null hypothesis**
 $H_0: \mu \geq 170$

2. Test Statistic

This is an example of a one-sample test of a mean when σ is known. Use this statistic to test the problem (for one sided or two sided the z statistic is the same)

$$\begin{aligned} Z_{\text{stat}} &= \frac{\bar{x} - \mu_0}{\sigma_{\bar{x}}} \\ &= \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \end{aligned}$$

3. Z score

α alpha	Two tailed $Z_{\alpha/2}$	One tailed Z_{α}
0.01	2.58	2.33
0.05	1.96	1.64
0.1	1.64	1.28

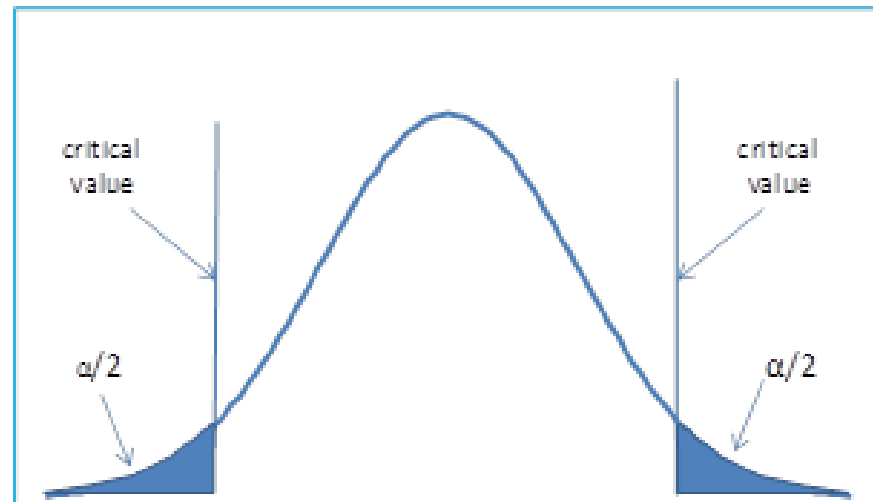
4. Conclusion

The decision is about the alternative hypothesis

We accept H_a or reject H_a .

To accept the alternative hypothesis we need to satisfy the following:

- Two sided
- $Z_{\text{stat}} < -Z_{\alpha/2}$ OR $Z_{\text{stat}} > Z_{\alpha/2}$ then we accept the H_a and we can claim its true
- Other wise we reject H_a



When its one sided

- If $\mu < \mu_0$

Then $Z_{\text{stat}} < -Z_{\alpha}$ so we can decide to accept H_a

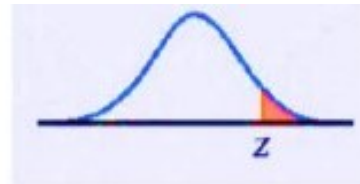
If not we reject H_a

- If $\mu > \mu_0$

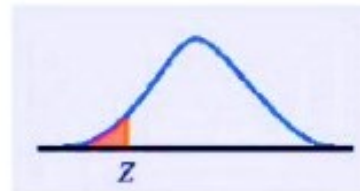
Then $Z_{\text{stat}} > Z_{\alpha}$ so we can decide to accept H_a

If not we reject H_a

$H_a: \mu > \mu_0$ is $P(Z \geq z)$



$H_a: \mu < \mu_0$ is $P(Z \leq z)$



The full example

In the 1970s, 20–29 year old men in the U.S. had a mean μ body weight of 170 pounds. Standard deviation $\sigma=40$. we collected a sample of $n = 64$, and mean of 185 . test whether mean body weight in the population now is more than 170 pounds.

$$(\alpha=0.01), \sigma_{\bar{x}} = \frac{40}{8} = 5$$

$$H_a: \mu > 170 \text{ (**one-sided test**)} \quad H_0: \mu \leq 170$$

- $Z \text{ stat} = 185 - 170 / 5 = 3$
- $Z\text{-score} = 2.33$
- $Z_{\text{test}} > Z_{\alpha}$ so we can decide to accept H_a and say that mean body weight in the population now is more than 170 pounds

Example

Researchers are interested in the mean age of •
a certain population That is distributed
normally with variance=20. If we took a
sample of 10 individuals with mean =27. Can
we conclude that the mean age of this
population less than 30 years? ($\alpha = 0.05$)

$$H_a: \mu < 30 \text{ (one-sided test)}$$

$$H_0: \mu \geq 30$$

$$Z_{\text{stat}} = \frac{27-30}{\sqrt{\frac{20}{10}}} = -2.12$$

Z-score = 1.64

$Z_{\text{stat}} < -Z_{\alpha}$ so we can decide
to accept H_a

that the mean age of this
population is less 30 years

Example

Researchers are interested in the mean age of •
a certain population That is distributed
normally with variance=20. If we took a
sample of 10 individuals with mean =32. Can
we conclude that the mean age of this
population differs than 30 years? ($\alpha = 0.05$)

$$H_a: \mu \neq 30 \text{ (two-sided test)}$$

$$H_0: \mu = 30$$

$$Z_{\text{stat}} = \frac{33 - 30}{\sqrt{\frac{20}{10}}} = 1.41$$

Z-score = 1.96

$Z_{\alpha/2} > Z_{\text{stat}} > -Z_{\alpha/2}$ so we can
decide to reject H_a

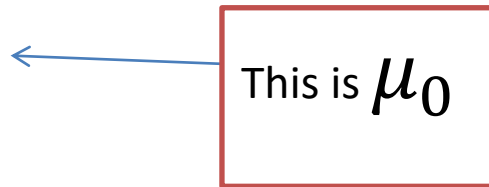
We don't have enough
evidence that the mean age of
this population differs than 30
years

Example

The problem: we want to test if the girls height average in J.U.S.T. is greater than 165 cm. We took a sample of 49 girls with average height of 167 and variance of 36 ($\alpha=0.1$)

$H_a: \mu > 165$ (**one-sided test**)

$H_0: \mu \leq 165$



This is μ_0

2. Test Statistic

This is an example of a one-sample t test of a mean when σ is unknown. Use this statistic to test the problem (for one sided or two sided the t statistic is the same)

$$t_{\text{stat}} = \frac{\bar{x} - \mu_0}{SE_{\bar{x}}}$$

$$= \frac{\bar{x} - \mu_0}{\frac{s \text{ or } \sigma}{\sqrt{n}}}$$

Example: t statistic

For the illustrative example,

Test if

$$H_a: \mu > 165 \text{ (one-sided test)}$$

$$H_0: \mu \leq 165$$

$\mu_0 = 165$, $s=6$ and $n = 49$. Therefore degrees of freedom= $n-1$

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{6}{\sqrt{49}} = .857$$

if sample mean is 167, then

$$t_{\text{stat}} = \frac{\bar{x} - \mu_0}{SE_{\bar{x}}} = \frac{167 - 165}{.857} = 2.334$$

Step 3:

D.f.= 48

One sided and we are using $\alpha=0.1$

$$t_{48, 0.1} = 1.282$$

Step 4:

$t_{stat} > t_{d.f., \alpha}$ then we accept the H_a and we can claim its true  $2.334 > 1.282$

Then we accept the claim that the girls height is greater 165